Tech Nation Visa — Global Talent (Exceptional Promise)

Criteria: Optional Criteria 3 — Product & Engineering Impact

Applicant: Odeyemi Olusegun Israel

OC3-2: UI Integrity Architecture, RBAC Enforcement, and Bybit-Class Attack Prevention

1. Executive Summary

I built: an enterprise compliance interface with cryptographic UI integrity safeguards, six-tier role-based access control, and AI decision explainability an end-to-end system I solely architected, built, and delivered to production. This work addresses a proven, high-impact threat: in February 2025, attackers at Bybit replaced a compliance approval interface with a tampered version, causing staff to unknowingly authorise a **$1.5 billion theft**.

2. UI Integrity Implementation

The file ui-integrity.ts provides four layered defences: (1) every screen component carries a unique cryptographic fingerprint; (2) approvals are signed against actual transaction data, not the displayed text; (3) the server independently validates the fingerprint before acting; and (4) a final confirmation is shown in a secure, isolated context. If an attacker replaces any part of the interface, the fingerprints will not match and the transaction is blocked before any approval can be given.

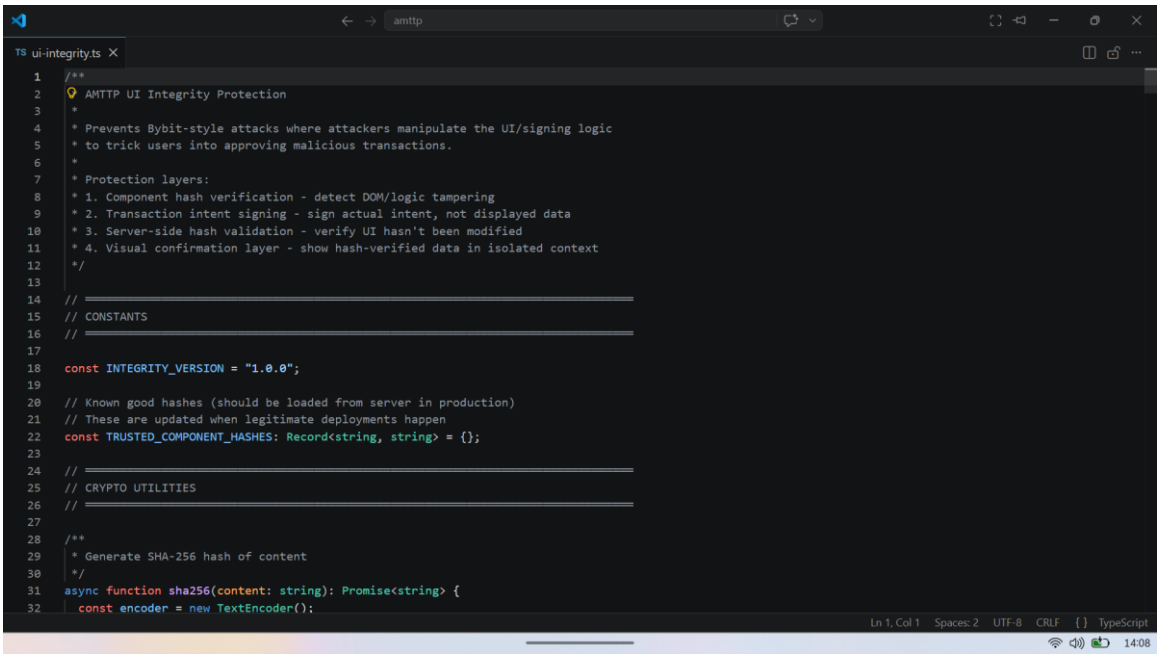

Figure 1: ui-integrity.ts lines 1–32 — header names the Bybit attack vector, lists four prevention layers, implements SHA-256 hashing via Web Crypto API. Any interface substitution breaks every hash.

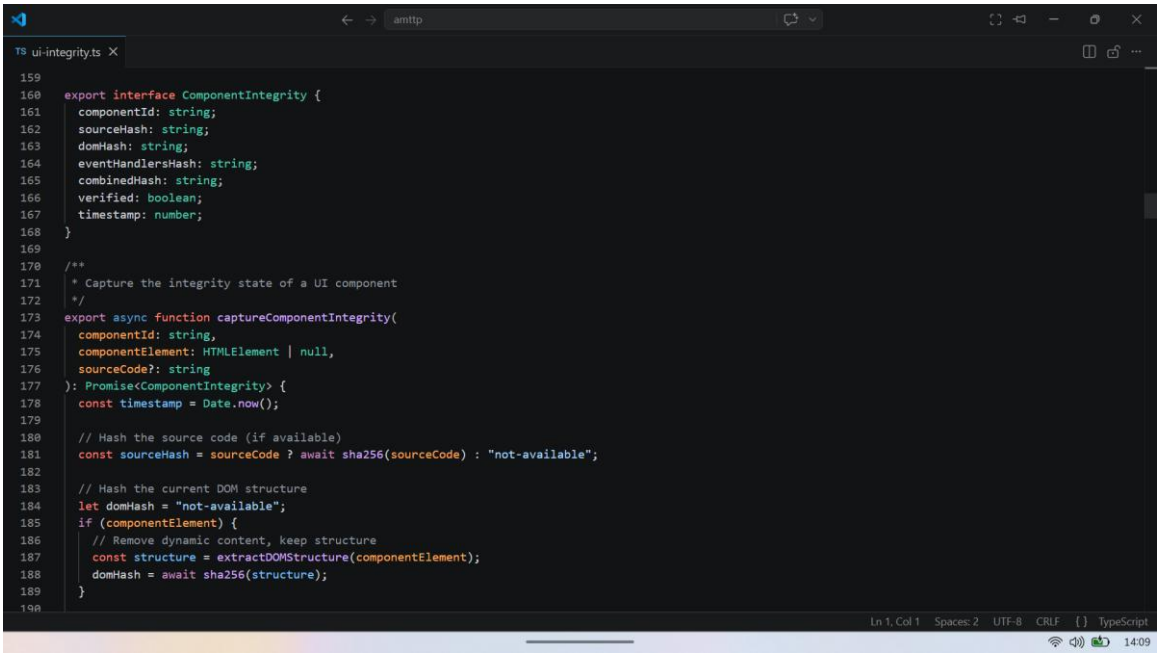

Figure 2: ui-integrity.ts lines 160–190 — four independent hashes (sourceHash, domHash, eventHandlersHash, combinedHash) per UI component. Hashing source code and live DOM independently detects the Bybit attack pattern.

3. Role-Based Access Control

Role	Label	Interface	Key Capabilities
R1	End User	Flutter (Focus Mode)	Own transactions only
R2	PEP End User	Flutter (Focus Mode)	Enhanced monitoring, restricted view
R3	Compliance Officer	Next.js (War Room)	All transactions, report generation
R4	Risk Analyst	Next.js (War Room)	Detection studio, policy editing
R5	Administrator	Next.js (War Room)	User management, enforcement actions
R6	Super Admin	Next.js (War Room)	Emergency override, full audit access

Each role is defined in a shared configuration file used by both the mobile app and the web dashboard. Access restrictions are enforced at the backend server level rather than on the visible interface, meaning they cannot be bypassed by manipulating what appears on screen.

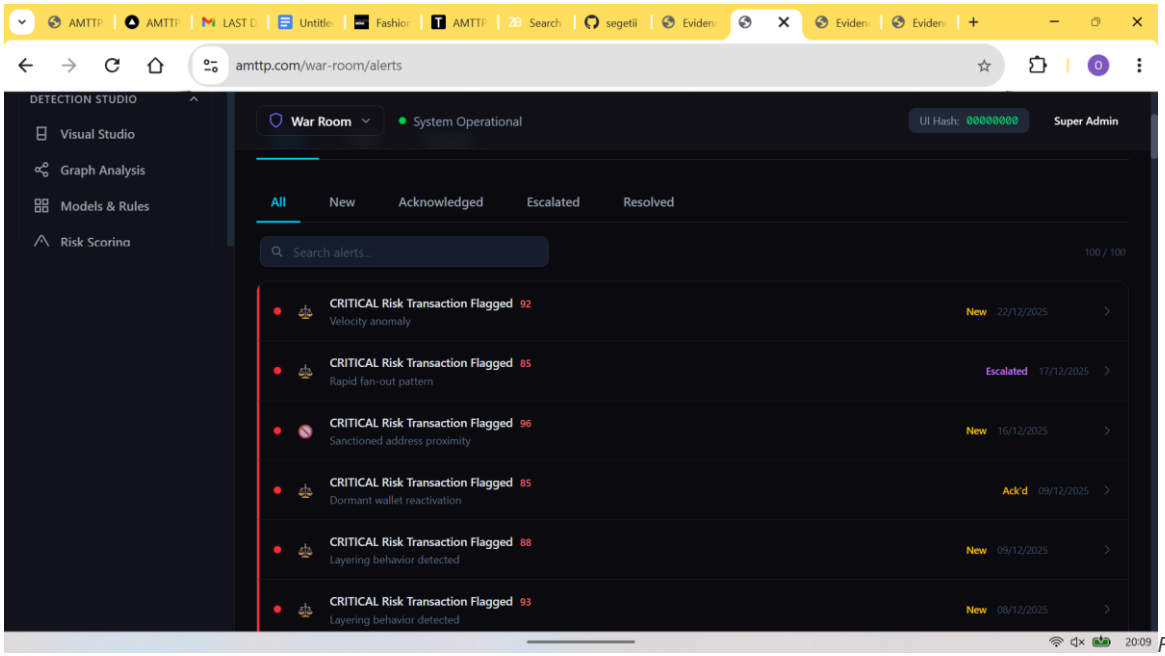

Figure 3: War Room Alerts panel (Compliance Officers and above, enforced server-side) 100 live critical alerts with severity scores and flag reasons. Role access is denied at the backend, so bypassing the UI restriction gains nothing.

4. Model Explainability and Audit Trail

If an officer receives a "CRITICAL — Block" decision with no reasoning, they cannot detect a manipulated model. The Explainability panel surfaces the contributing factors and their weighting for every automated decision, satisfying FCA/MiCA audit trail requirements.

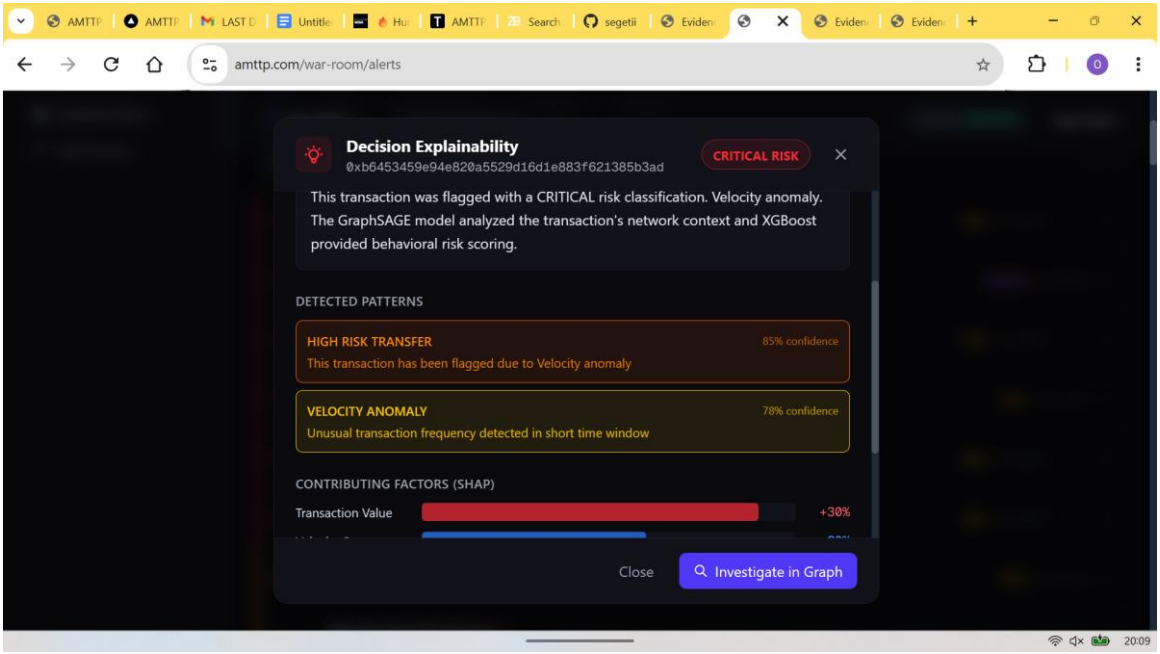

Figure 4: Decision Explainability panel for a CRITICAL alert — contributing factors and their weights: High Risk Transfer 85%, Velocity Anomaly 78%, Sanctioned Address Proximity 65%, with direct link to the graph network view. Satisfies FCA/MiCA audit trail requirements without additional tooling.